

Simulation 3

DCEN + EducationLibre

European ENR Coalition
with BitTorrent P2P Education model

- 15 member countries - DCEN-ENR Coalition
- EducationLibre: peer-to-peer credit system
- Feynman method + Non-violent communication
 - Study groups of 15 max - Social contract
 - Teacher/student validation - Sub-subjects

1. EducationLibre - Analysis of the source model

EducationLibre (github.com/nadnone/EducationLibre) is an open source project proposing a parallel education system inspired by the BitTorrent protocol. It functions as a patch on the classical education system, not as its replacement. Its philosophy is directly compatible with the DCEN model.

1.1 Fundamental principles of EducationLibre

BitTorrent model applied to education

As in BitTorrent, each member is simultaneously a client and server of knowledge. A student who masters trigonometry can teach it to others, after validation by a certified teacher. Subjects are subdivided into granularly teachable sub-branches - exactly like torrent files.

Incentive credit system - non-exchangeable, transfer only

A credit sub-system incentivises members not just to learn but also to teach what they master. Credits are NEVER exchanged between members - they are transferred only to the managing cooperative, association or foundation via a mobile application. This guarantees that credits remain a measure of internal social reciprocity, never a peer-to-peer currency.

Study groups of 15 people maximum - virtual, in-person or offline

The number 15 is deliberate: large enough for diversity, small enough for personalised teaching and a cohesive group dynamic. Groups can function in three ways: online via an app, in-person with connection, or offline with a paper notebook synchronised at the next connection. A single smartphone in a group of 15 is sufficient to synchronise everyone.

Explicit social contract per group

Each study group negotiates its own social contract within the framework of the state contract. This subsidiarity of the social contract is directly homologous to the reversed cantonal governance of the DCEN model.

Non-violent communication (NVC)

The model integrates Marshall Rosenberg's NVC as the internal communication protocol. It is the social lubricant that allows heterogeneous groups - unemployed, sick, retired, working - to coexist and learn together.

Validation by certified teacher

Before teaching a sub-subject, a student is assessed by a certified teacher. This validated seeding mechanism guarantees network quality while preserving the horizontality of the system.

1.2 EducationLibre x DCEN synergy

DCEN component	EducationLibre contribution	Synergy
Green SSE (cooperatives)	Internal P2P training	The SSE becomes self-learning without external consultants
Cantonal governance	Local study groups	Citizens able to understand and vote on budgets
4-day week	Time freed for study	Free time becomes collective epistemic capital
Inclusion unemployed/sick	'People in difficulty'	EducationLibre explicitly targets this public in its objectives
Sovereign fund	Funds infrastructure	Credit server + API + mobile app + certified teacher bootstrapping

Rural areas / digital divide	Offline mode	In-person groups with paper notebook, synchronised at next connection
Virtuous cycle	Credits = internal reciprocity	Knowledge circulates according to its own logic - independent of DCEN currency

1.3 Fundamental principle: two distinct and watertight systems

NON-NEGOTIABLE PRINCIPLE: EducationLibre credits and the real currency of the DCEN system are two separate and watertight systems. They are not convertible into one another. This separation is not a technical constraint - it is a founding philosophical decision that protects the integrity of both systems.

Why EducationLibre credits must NOT be convertible into money

Merging the two systems would destroy intrinsic motivation. If teaching trigonometry earns money, one teaches to earn - not to share. Psychology research (Deci & Ryan, Self-Determination Theory) documents this: introducing an extrinsic reward into an already intrinsically motivated activity destroys it. EducationLibre credits are a currency of belonging and reciprocity - they mean 'you contributed to the network', not 'you earned X euros'.

What the DCEN sovereign fund finances within EducationLibre

The sovereign fund finances ONLY the material conditions of the network: digital infrastructure (servers, API, mobile application), paid validator teacher posts, premises for in-person groups, internet access for precarious persons, and NVC training. It does not finance the credits themselves and does not monetise them. Bootstrapping mechanism: external persons or organisations (companies, municipalities, individuals) can BUY credits from the managing cooperative. This money is used solely to remunerate certified institutional teachers to start the network and validate the first teachers. This is the BitTorrent bootstrapping mechanism: professional seeders are created so the network can then function autonomously without permanent monetary injection.

The correct model: electricity vs conversations

The relationship between DCEN and EducationLibre is that of the electricity grid and the conversations that take place within it. The DCEN supplies the current (infrastructure, funding, governance). EducationLibre decides what to do with that current (knowledge, reciprocity, social bonds). Neither interferes with the logic of the other. Cutting the power does not stop conversations - they continue in person. And conversations do not need to know where the power comes from.

What EducationLibre credits actually represent

Credits are a social memory of contribution to the network - close to Wikipedia or StackOverflow reputation points, not a currency. They allow access to teaching from other members (reciprocity logic), signal one's level of mastery of a sub-subject (validation logic), and measure network activity (governance logic). They have no exchange rate, no market, and cannot be accumulated indefinitely.

Mobile application and offline mode: the technical infrastructure

The mobile application is the point of contact between members and the managing cooperative. It records credits (transferred to the cooperative, never between members), manages virtual study groups, and synchronises offline groups when connectivity returns. Offline mode: a rural group can function with a paper notebook for weeks. At the next connection of any member, the application synchronises all sessions and updates the credits of all participants. Access to knowledge never depends on connection quality.

Bootstrapping mechanism: credit purchase by external actors

To start the EducationLibre network, the classic chicken-and-egg problem arises: no seeders without members, no members without seeders. Solution: external actors (SSE companies, municipalities, sovereign fund, engaged individuals) can BUY credits from the managing cooperative. This money does not create speculable internal currency - it directly finances certified institutional teacher posts charged with validating the first network teachers and teaching the first sub-subjects. Once the network is self-sustaining, external monetary injection can gradually reduce. This is exactly the cooperative model: start-up social capital for a structure that then becomes autonomous.

2. Parameters of Simulation 3

Simulation 3 takes the base of simulation 2 (European coalition of 15 countries) and integrates the EducationLibre model as the P2P education infrastructure. EducationLibre-specific parameters are added to the DCEN-ENR parameters.

2.1 Parameters inherited from Simulation 2 (EU coalition)

Parameter	Value	Source
Coalition member countries	15	Sim. 2 - critical mass
Mutualised electricity grid	55%	Sim. 2 - energy optimum
European common sovereign fund	40%	Sim. 2 - capitalisation
DCEN model adoption	60%	Sim. 2 - coherence
Common fiscal blockchain	60%	Sim. 2 - anti-evasion
Coordinated fossil phase-out	50%	Sim. 2 - ENR transition

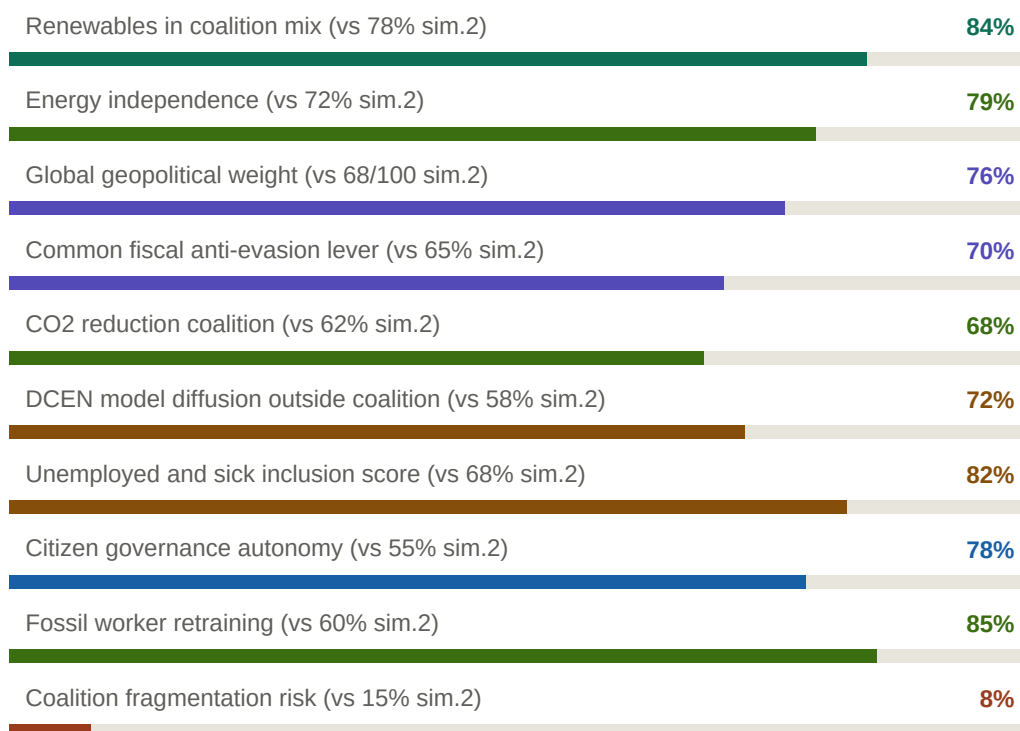
2.2 EducationLibre-specific parameters (new)

EducationLibre parameter	Sim. value	Justification
EducationLibre adoption rate in coalition	65%	Gradual start - pilot associations in each member country
Active study groups (per 100k inhabitants)	12	Target: 1 group per approximately 8,000 inhabitants
Sovereign fund share to EL infrastructure	8%	Servers, API, mobile app, offline mode, certified teacher bootstrapping
Validated teacher / student ratio	1/6	1 validated teacher for 6 students on average
Sub-subjects registered in EU network	850+	From trigonometry to renewable energy
NVC integration in group contracts	80%	NVC training funded by the cantonal sovereign fund
Feynman method as pedagogical standard	Yes	Recommended by EducationLibre - validates deep understanding

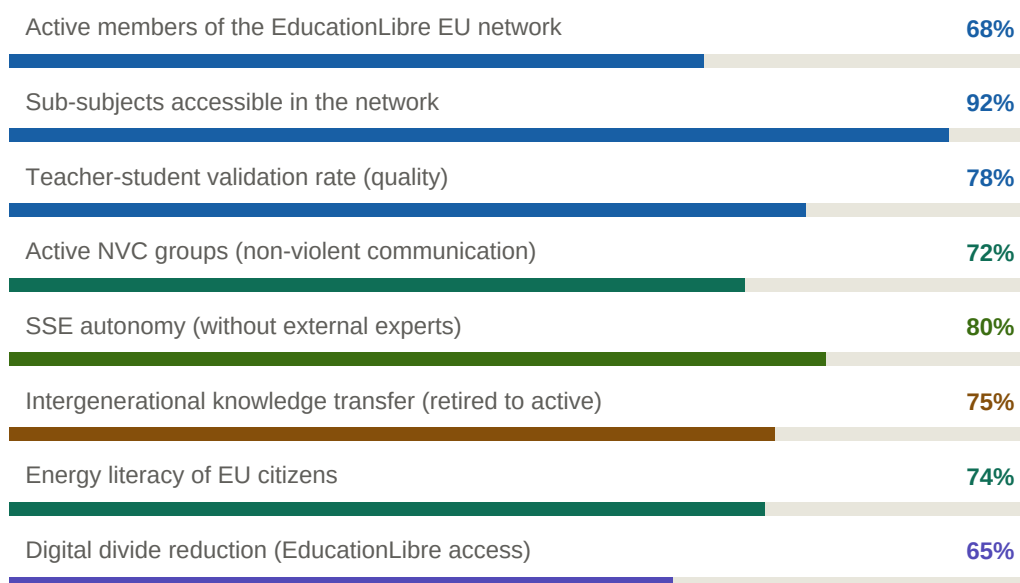
3. Results of Simulation 3

The integration of EducationLibre into the DCEN-ENR European coalition produces multiplier effects on virtually all indicators from previous simulations, and creates new ones impossible to reach without this layer.

3.1 DCEN indicators enriched by EducationLibre



3.2 EducationLibre-specific indicators (new)



4. Comparison of the three simulations

Indicator	Sim.1 Nat. ENR	Sim.2 EU	Sim.3 EU+EL	EL gain
Renewables mix	72%	78%	84%	+6 pts
CO2 reduction	58%	62%	68%	+6 pts
Social inclusion	68%	68%	82%	+14 pts
DCEN diffusion	Local	58%	72%	+14 pts
Citizen autonomy	45%	55%	78%	+23 pts
Fossil retraining	48%	60%	85%	+25 pts
Fragmentation risk	18%	15%	8%	-7 pts
Governance without experts	30%	40%	80%	+40 pts
Intergenerational social bond	N/A	N/A	75%	New
Energy literacy	30%	45%	74%	+29 pts

EducationLibre produces the largest gains precisely where simulations 1 and 2 were weakest: citizen autonomy, fossil worker retraining, governance without dependence on experts, and intergenerational social bonds. These are exactly the points the DCEN model promised without having the mechanism to deliver.

5. Specific EducationLibre mechanism in the DCEN-EU context

5.1 How the EducationLibre network deploys within the coalition

Phase	EducationLibre action	DCEN link	Timeline
Bootstrapping	Pilot associations in 3 cities per member country	Cantonal sovereign fund financing (8%)	Year 1-2
Growth	Common EU credit server, open API	Mutualised DCEN blockchain infrastructure	Year 2-5
Maturity	850+ sub-subjects, 15 EU languages, 68% active members	Self-governing SSE, CCHI understood by members	Year 5-15
Diffusion	Model exported to Africa, Asia, Latin America	DCEN standard taught by EducationLibre in partner countries	Year 10-25

5.2 Strategic sub-subjects for the DCEN

Strategic subject	Why it is key for the DCEN
Cooperative governance and SSE accounting	Allows SSE members to understand their own organisation without external audit.
Progressive taxation and blockchain (basics)	Citizens understand the system that taxes them - maximum democratic legitimacy.
Renewable energy and energy sobriety	Energy literacy to accelerate adoption of ENR cooperatives.
Labour law and cooperative contracts	Legal autonomy for workers retrained from fossil sectors.
Non-violent communication (NVC)	Basic protocol for all study groups and citizen assemblies.
Preventive health and CCHI	Understanding one's cooperative health fund to govern it better.
History of cooperative movements	Historical context to anchor the DCEN model in a living tradition.
Pedagogy and the Feynman method	Training the trainers - every member a potential teacher.

6. Specific advantages of the EducationLibre integration

The SSE moves from dependent to autonomous

In simulations 1 and 2, the SSE depended on external experts for its governance. EducationLibre trains members from within: cooperative accounting, labour law, and project management become network sub-subjects. The SSE no longer needs a consulting firm - it knows how to govern itself.

Fossil workers retrain through knowledge, not unemployment

Retraining rises from 60% (sim.2) to 85% (sim.3). EducationLibre allows miners, oil and gas workers to learn renewable energy trades via local study groups, validated by certified technicians. A just transition is no longer a slogan - it is an operational mechanism.

The European coalition acquires epistemic resilience

A coalition that shares not only its electricity grid but also its knowledge network is much harder to fragment. If a country leaves the coalition, its citizens retain the knowledge acquired via EducationLibre. Cohesion is epistemic, not merely institutional.

The reciprocity network creates a cross-border epistemic bond

EducationLibre credits - not convertible into real money - circulate between member countries according to their own reciprocity logic. A Danish student teaches offshore wind to a Portuguese student who teaches them solar energy. This is not a financial transaction: it is a knowledge exchange governed by EducationLibre's internal credit system. The two systems coexist without contaminating each other.

The Feynman method guarantees quality without centralisation

By requiring each student to be able to re-explain what they have learned simply (Feynman method), EducationLibre guarantees deep understanding, not superficial memorisation. Combined with the DCEN, this means citizens who truly understand their economy and can defend it.

Inclusion of people in difficulty is structural

EducationLibre explicitly targets 'people in difficulty of all kinds within the limits of their cognitive abilities'. This inclusion is in the DNA of the system, not in its secondary objectives. Combined with NVC and groups of 15 max, it becomes operational, not merely declarative.

7. Specific risks and limits

Quality not guaranteed without sufficient validated teachers

The system relies on validation by qualified teachers. If the ratio of validated teachers to students becomes too unbalanced (below 1/10), quality collapses and the network degrades like a poorly seeded torrent. The sovereign fund must finance professional 'super-seeder' posts.

Linguistic fragmentation in the EU coalition

27 languages in Europe. Sub-subjects must be available in all languages to guarantee equal access. P2P translation (on the Wikipedia model) is a solution but takes time. Risk of a two-speed coalition depending on the dominant language.

Internal credit system manipulation

A credit system can be manipulated: members exchanging credits without genuinely teaching, ghost groups, surface-level teaching to accumulate. Feynman validation and NVC are safeguards but not perfect solutions. Important note: the fact that credits are NOT convertible into real money considerably reduces the incentive to cheat - there is no financial gain from cheating in a system of pure social reciprocity.

Resistance from institutional education systems

EducationLibre explicitly positions itself as a 'patch', not a replacement. But teachers' unions and education ministries might see in this system a threat to their monopoly. The coalition must negotiate explicit complementarity agreements, not competition.

Residual digital divide - attenuated by offline mode

The offline mode of EducationLibre significantly reduces this risk: in-person groups function with a paper notebook and synchronise credits at the next connection of any member. A single person with a smartphone in a group of 15 is sufficient. The residual risk remains for non-readers or those with significant cognitive impairments - NVC and groups of 15 max are partial responses but not sufficient for 100% of situations.

Cognitive load on members

Teaching AND learning ON TOP of working (even 4 days) is cognitively demanding. Social recognition within the EducationLibre network (via internal reciprocity credits, not convertible into money) must be sufficiently rewarding for the contribution to feel enriching rather than burdensome. The 4-day week is key here: it frees up real time for learning.

8. Conclusion - The complete DCEN v3 model

Simulation 3 reveals that EducationLibre is not simply an additional building block of the DCEN - it is the mechanism that transforms an economic model into a learning civilisation. Without EducationLibre, the DCEN redistributes wealth. With EducationLibre, it also redistributes knowledge. And as Piketty has shown, knowledge is the form of capital most equitably distributed - and the hardest to confiscate.

Complete DCEN v3 model formula

Global blockchain (Zucman) + Reversed cantonal taxation (Saez/Piketty) + Citizens sovereign fund + Green SSE + CCHI + 4-day week + EU ENR Coalition (15 countries) + EducationLibre BitTorrent (NVC + Feynman + credits) = Plural, democratic, sober AND learning economy.

Complete DCEN v3 feasibility matrix

Building block	Technical feasibility	Political feasibility	Timeline
Green SSE + local EducationLibre	High	High	Immediate
Reversed cantonal taxation	High	High	2-5 years
EducationLibre EU (credit network)	High	High	3-7 years
Generalised 4-day week	Medium	High	3-7 years
EU ENR Coalition 15 countries	High	Medium	7-15 years
Sovereign fund + CCHI	High	Medium	7-15 years
Common EU fiscal blockchain	High	Medium	5-12 years
CCHI (Swiss KVG reform)	High	Low	10-15 years
EducationLibre worldwide (export)	High	Medium	15-25 years
Global wealth blockchain	High	Low	15-25 years

Main sources: EducationLibre - github.com/nadnone/EducationLibre (CC BY 4.0) | T. Piketty - Capital in the Twenty-First Century | E. Saez - Income Inequality in the US | G. Zucman - The Hidden Wealth of Nations | R. Feynman - Pedagogy through explanation | M. Rosenberg - Nonviolent Communication | OECD - Tax Agreement 2021 | BitTorrent Protocol - Cohen, 2001 | Deci & Ryan - Self-Determination Theory (1985)